



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

26-03-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B	B,C	A,C,D	A,B,C	C,D	A		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	A	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.50	20.30	0.05	1.53	1.15 or 1.16	1.00	0.80	152.33 or 152.35

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	B,D	A,B,C	A,C,D	A,C,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	D	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	9.00	4.00	2.00	18.00	2.50	1.30	3.73 or 3.74	1.88 or 1.89

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	B,C	A,B,D	A,C,D	B,C	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	3.00	4.00	4.00	8.00	2.00	8.00	5.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

For C.O.M. position $\Rightarrow m_1 x_1 = m_2 x_2$

where m_1 & m_2 are masses of left & right part of the rod end

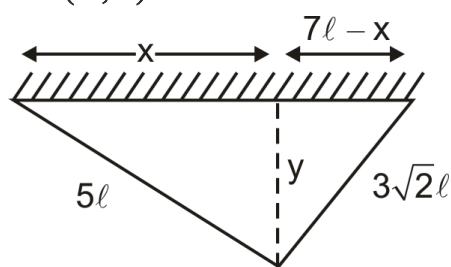
x_1 & x_2 are their respective position of center of mass.

simultaneously, $m_1 x_1 \omega = m_2 x_2 \omega$ (since C.M is at rest)

and $I_1 \omega \neq I_2 \omega$

and $\frac{1}{2} I_1 \omega^2 \neq \frac{1}{2} I_2 \omega^2$.

2. Ans (B,C)



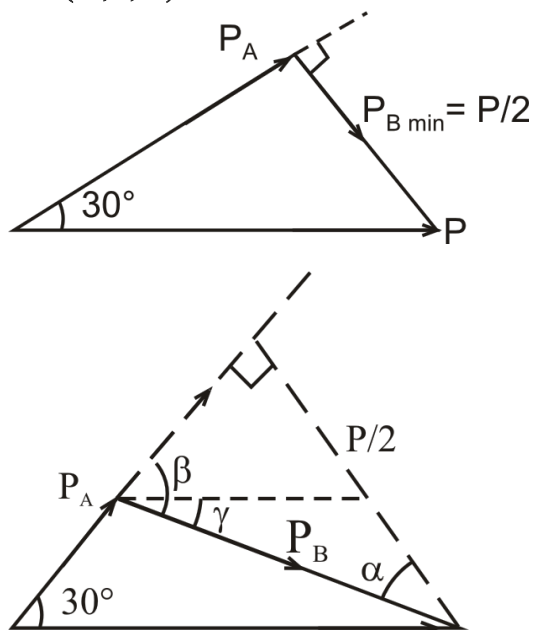
$$y^2 = (5l)^2 - x^2$$

$$= (3\sqrt{2}l)^2 - (7l - x)^2$$

$$x = 4l, y = 3l$$

$$T = 2\pi \sqrt{\frac{3l}{g}}$$

3. Ans (A,C,D)



P_B will be minimum when P_A is $\perp P_B$.

$$\text{but } P_B = \frac{2}{\sqrt{3}} \times P_{B \min} = \frac{2}{\sqrt{3}} \times \frac{P}{2} = \frac{P}{\sqrt{3}}$$

$$\cos \alpha = \frac{P/2}{P_B} = \frac{P/2}{P/\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\alpha = 30^\circ$$

so angle between $\vec{P}_A$ & $\vec{P}_B$ is 60°

$$P_A \cos 30^\circ + P_B \cos \gamma = P, \gamma = \beta - 30^\circ = 30^\circ$$

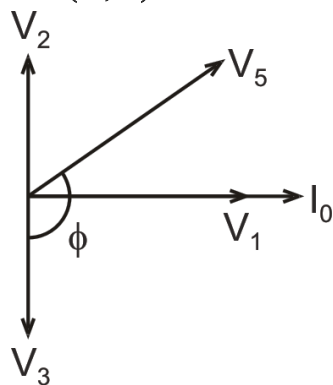
$$P_A \cos 30^\circ + \frac{P}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} = P$$

$$P_A \times \frac{\sqrt{3}}{2} = \frac{P}{2} \Rightarrow P_A = \frac{P}{\sqrt{3}}$$

$$\text{Loss in K.E.} = \frac{P^2}{2m} - \left[\frac{P^2}{3 \times 2m} + \frac{P^2}{3 \times 2m} \right]$$

$$= P^2/6m.$$

5. Ans (C,D)



$$e = e_1 + e_2 = 200 \sin \left(\omega t + \frac{\pi}{3} \right)$$

$$e_0 = 200 \text{ volt}$$

$$V_5 = \frac{200}{\sqrt{2}} = 100\sqrt{2}$$

$$z = \sqrt{(500)^2 + (500)^2} = 500\sqrt{2}$$

$$i_{\text{rms}} = \frac{100\sqrt{2}}{500\sqrt{2}} = \frac{1}{5}$$

$$V_1 = 500 \times \frac{1}{5} = 100 \text{ volt}$$

$$V_2 = 900 \times \frac{1}{5} = 180 \text{ volt}$$

6. Ans (A)

$$\text{Potential at 'C'} = \frac{kQ}{R} = \frac{\sigma R}{2\epsilon_0}$$

Flat surface of hemispherical shell is on equipotential surface.

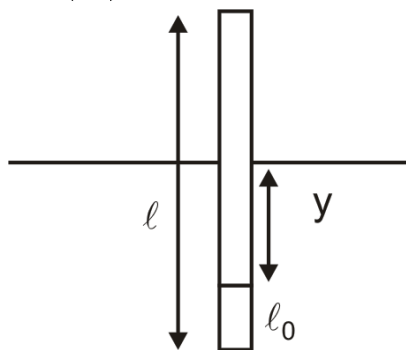
PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (D)

$$Q \text{ value} = (\text{B.E.})_{\text{products}} - (\text{B.E.})_{\text{reactants}}$$

9. Ans (A)



$$\ell_0 A \times 2.7 + \ell A \times 0.8 = (\ell_0 + y)A$$

$$\ell_0(2.7 - 1) + 0.8\ell = y$$

$$y = 1.7\ell_0 + 0.8\ell$$

$$y = 1.7 + 0.8\ell$$

$$\text{at } t = 5 \text{ minute; } \ell = 9.5 \text{ cm } y = 9.3$$

$$\text{at } t = 10 \text{ minute; } \ell = 9 \text{ cm } y = 8.9$$

$$\text{at } t = 15 \text{ minute; } \ell = 8.5 \text{ cm } y = 8.5$$

$$\text{at } t = 20 \text{ minute; } \ell = 8 \text{ cm } y = 8.1 \text{ which is not}$$

$$\text{possible} \Rightarrow y = 8.5$$

$$T = 2\pi \sqrt{\frac{m}{A\rho g}}$$

$$m_0 = k(2.7 + 8) = 10.7 \text{ k}$$

$$m_5 = k(2.7 + 7.6) = 10.3 \text{ k}$$

$$m_{10} = k(2.7 + 7.2) = 9.9 \text{ k}$$

$$m_{15} = k(2.7 + 6.8) = 9.5 \text{ k}$$

$$m_{20} = k(2.7 + 6.8) = 9.5 \text{ k}$$

$$T_0 \propto \sqrt{10.7}$$

$$T_5 \propto \sqrt{10.3} \Rightarrow \sqrt{\frac{10.3}{10.7}} T_0$$

$$T_{10} = \sqrt{\frac{9.9}{10.7}} T_0$$

$$T_{15} = \sqrt{\frac{9.5}{10.7}} T_0$$

$$T_{20} = \sqrt{\frac{9.5}{10.7}} T_0$$

10. Ans (C)

(I) $0 = 100V + 50(V + 3)$ – conservation of linear momentum.

work done by man

$$= \frac{1}{2} (100)(1^2) + \frac{1}{2} (50)(2^2) = 150 \text{ J.}$$

$$(II) W_N + W_{Mg} = \Delta K.E.$$

$$W_N - \frac{1}{2} (2) (10)^2 = \frac{1}{2} (2) [(5)^2 - (15)^2]$$

$$W_N = -100 \text{ J.}$$

(III) Unstable equilibrium points are

$$x = 10 \text{ m, } 30 \text{ m.}$$

stable equilibrium point is $x = 20 \text{ m.}$

$$\text{so } U_i = 0,$$

$$\text{and } U_f = -\frac{1}{100} (10^4) = -100 \text{ J.}$$

$$W_{\text{conservative}} = U_i - U_f = 100 \text{ J.}$$

$$(IV) W_{\text{ext}} = U_f - U_i = mg \left(\frac{5R}{8} - \frac{3R}{8} \right) = \frac{mgR}{4} = 100 \text{ J.}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (1.50)

$$v_{\text{rms}}^2 = \frac{\int v^2 dN}{\int dN} = \frac{\int v^2 \frac{dN}{dv} \cdot dv}{\int \frac{dN}{dv} \cdot dv} = \frac{\int_0^3 v^2 \left(\frac{-5}{3}v + 5 \right) dv}{\int_0^3 \left(\frac{-5}{3}v + 5 \right) dv} = 1.50$$

2. Ans (20.30)

Let final pressure on both side of W_2 is P

$$P' \left(\frac{V_0}{6} \right)^\gamma = P_0 \left(\frac{5V_0}{6} \right)^\gamma$$

$$P' = 5\sqrt{5} P_0$$

$$\text{Work done by gas} = \frac{5\sqrt{5} P_0 \left(\frac{V_0}{6} \right) - P_0 \cdot \frac{5V_0}{6}}{\gamma - 1}$$

$$= \frac{5P_0 V_0}{3} (\sqrt{5} - 1)$$

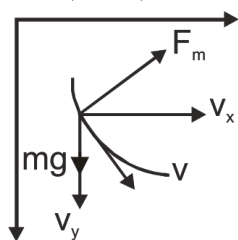
$$\Delta U = nc_v \Delta T = n \left(\frac{R}{\gamma - 1} \right) \Delta T = 2nR \Delta T$$

$$= 2 \left(P' \cdot \frac{5V_0}{6} - \frac{P_0 V_0}{6} \right) = 2 \left(\frac{5\sqrt{5} P_0 \cdot 5V_0}{6} - \frac{P_0 V_0}{6} \right)$$

$$= \frac{P_0 V_0}{3} (25\sqrt{5} - 1)$$

$$\Delta Q = \Delta U + \Delta W = 2P_0 V_0 (5\sqrt{5} - 1) = 20.3 P_0 V_0$$

3. Ans (0.05)



$$mg - q B v_x = m \frac{dv_y}{dt} \quad \dots (1)$$

$$q B v_y = m \frac{dv_x}{dt}$$

Differentiate equation (1) w.r.t. time.

$$0 - q B \frac{dv_x}{dt} = m \frac{d^2 v_y}{dt^2}$$

$$\frac{d^2 v_y}{dt^2} = \frac{-q^2 B^2}{m^2} v_y = -\omega^2 v_y$$

$$v_y = (v_y)_0 \sin \omega t$$

$$\frac{dv_y}{dt} = (v_y)_0 \omega \cos \omega t$$

$$\text{at } t = 0, \frac{dv_y}{dt} = g, \text{ so } (v_y)_0 = g/\omega$$

Now from equation (1)

$$v_x = \frac{mg}{qB} (1 - \cos \omega t)$$

$$\text{so } v_y = 0, \text{ at } \omega t = \pi, t = \frac{\pi}{\omega}$$

$$v_x = \frac{mg}{qB} [1 - (-1)] = \frac{2mg}{qB}$$

From work energy theorem

$$mg_y = \frac{1}{2} mv_x^2 \Rightarrow y = 0.05 \text{ m}$$

4. Ans (1.53)

$$\text{Pitch of the screw gauge} = \frac{5}{10} = 0.5 \text{ mm.}$$

$$\text{Least count} = \frac{\text{pitch}}{100} = \frac{0.5}{100} = 0.005 \text{ mm}$$

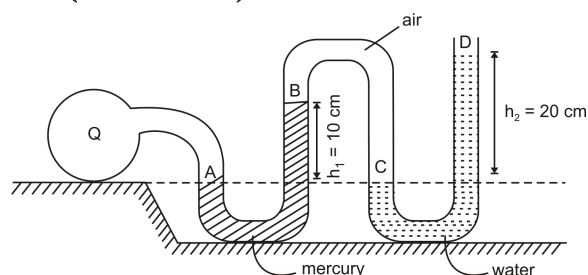
$$\text{Pitch scale reading} = \text{pitch} \times N = 0.5 \times 3 = 1.5 \text{ mm.}$$

$$\text{Difference of the cap reading} = \text{Final} - \text{Initial} = 6.$$

$$\text{Circular scale reading} = \text{Difference} \times \text{L.C.} = 0.03 \text{ mm.}$$

$$\text{Diameter} = (1.5 + 0.03) \text{ mm} = 1.53 \text{ mm.}$$

5. Ans (1.15 or 1.16)



$$P_C = P_D + \rho_2 g h_2 = P_0 + \rho_2 g h_2$$

$$P_A = P_B + \rho_1 g h_1$$

and $P_B = P_C$ (neglecting the density of air as compared to water or mercury).

$$P_A = P_0 + \rho_2 g h_2 + \rho_1 g h_1$$

$$= 10^5 + 10 (13.8 \times 0.1 + 1 \times 0.2) \times 10^3$$

$$= 10^5 + 10^4 (1.38 + 0.2) = 1.16 \times 10^5 \text{ N/m}^2.$$

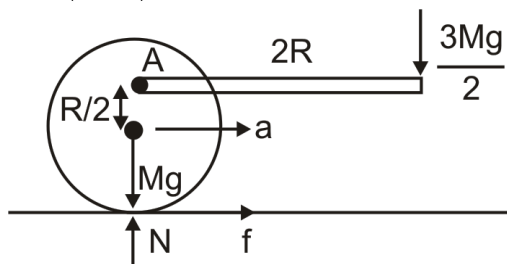
$$= 0.116 \text{ MN/m}^2$$

$$= 0.12 \text{ MN/m}^2$$

6. Ans (1.00)

$$H_{\text{app}} = \frac{H}{\mu} \left(\frac{\cos r}{\cos i} \right)^3 = \frac{9}{\sqrt{3}} \times \left(\frac{1}{\sqrt{3}} \right)^3 = 1 \text{ m}$$

7. Ans (0.80)



taking disc & rod as whole system.

$$\frac{3Mg}{2} + Mg = N$$

$$N = \frac{5Mg}{2}$$

torque about centre of disc

$$\frac{3Mg}{2} \times 2R - f \times R = \frac{MR^2}{2} \times \alpha$$

$$3Mg - f = \frac{MR\alpha}{2}, a = R\alpha.$$

$$3Mg - f = \frac{Ma}{2}, f = Ma$$

$$3Mg - Ma = \frac{Ma}{2}$$

$$3Mg = \frac{3Ma}{2}$$

$$a = 2g.$$

$$f = Ma = 2Mg$$

$$\mu N = 2Mg$$

$$\mu \times \frac{5Mg}{2} = 2Mg$$

$$\mu = 4/5 = 0.8$$

8. Ans (152.33 or 152.35)

$$\sin i = \frac{4}{3} \sin r$$

$$r = 37^\circ$$

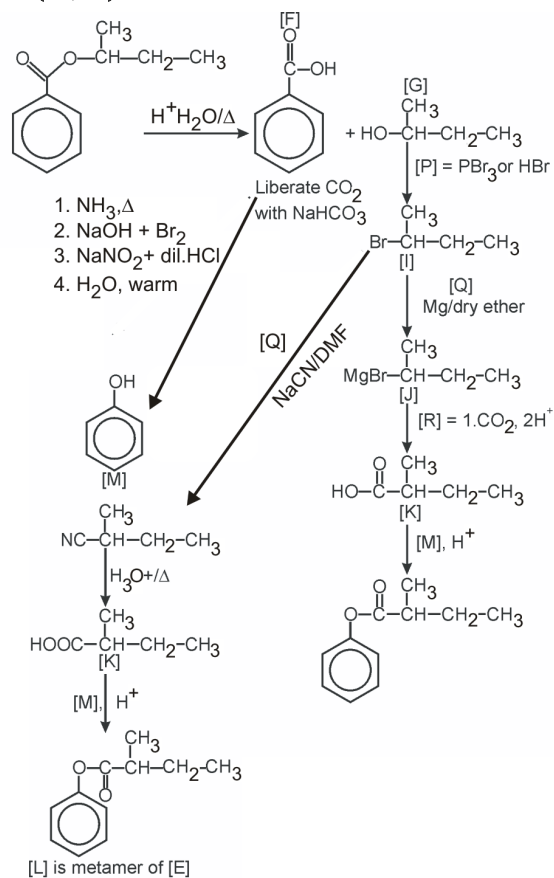
$$2\mu t \cos r = \lambda/2$$

$$t = \frac{\lambda}{4\mu \cos r} = 97.50 \text{ nm.}$$

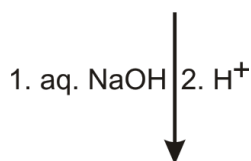
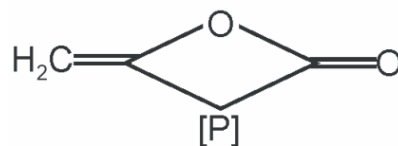
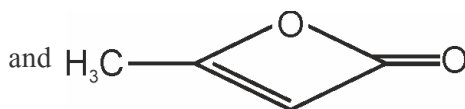
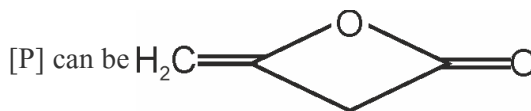
PART-2 : CHEMISTRY

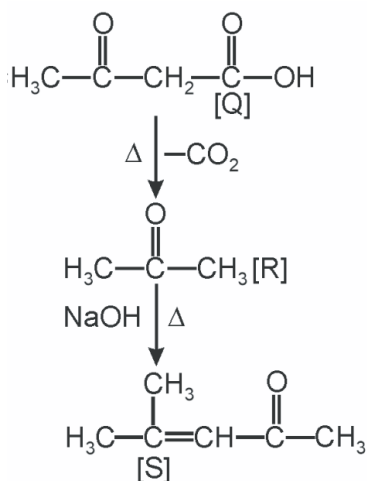
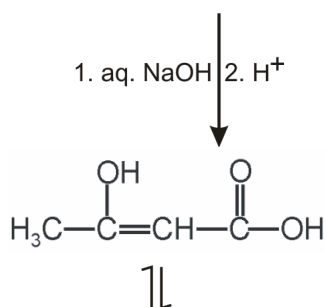
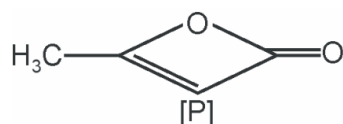
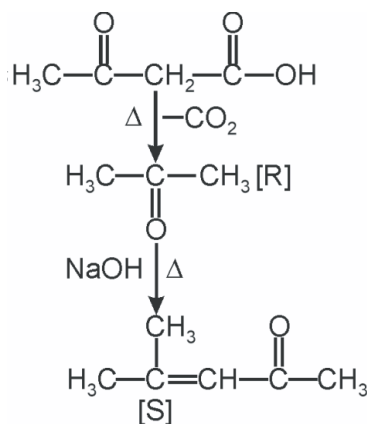
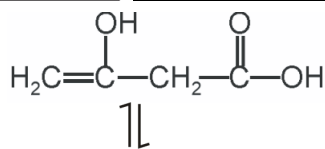
SECTION-I (i)

1. Ans (C,D)



2. Ans (B,D)





4. **Ans (A,C,D)**

In solid state dihedral angle of H_2O_2 is approx 90.2° at 110 K different from its gaseous state which is approx 111.5° .

5. **Ans (A,C,D)**

(A) is correct because number of ions in 1 ml decreases

(B) is incorrect as H_2O is oxidised at anode

(C) Conductance depends upon cell constant and concentration

(D) To calculate limiting equivalent conductivity Kohlrausch law is used.

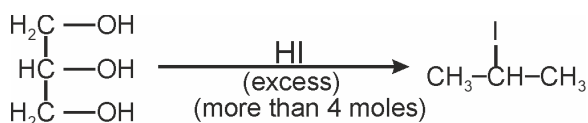
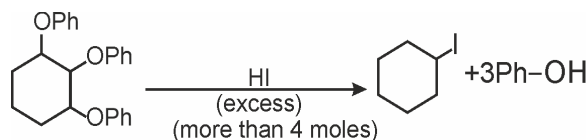
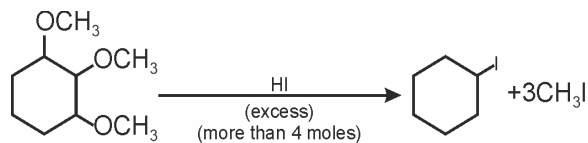
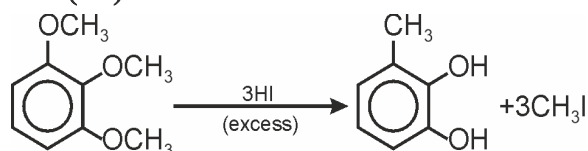
6. **Ans (A,B)**

Theory based.

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (D)**



8. **Ans (D)**

Sucrose $\xrightarrow{\text{Hydrolysis}}$ α -D-glucose + β -D-fructose

Lactose $\xrightarrow{\text{Hydrolysis}}$ β -D-glucose + β -D-galactose

Amylose $\xrightarrow{\text{Hydrolysis}}$ α -D-glucose + α -D-glucose +

Mannose cannot be hydrolysed.

Lactose and mannose can reduce Fehlin's solution due to presence of hemiacetal and aldehyde group.

10. Ans (A)

(I) $4\pi^2 R^2$ graph with r have $(n - \ell - 1)$

intersection point with x-axis except origin.

Hence I $\rightarrow$ P, T

(II) Angular probability depends upon θ and ϕ

for all p and d-orbitals. II $\rightarrow$ P, T, S.

$$\text{(III)} = \frac{2 \times \text{KE}_{2,3}}{3 \times \text{PE}_{4,2}} = \frac{2 \times 13.6 \times \frac{3^2}{2^2}}{3 \times \left(-2 \times 13.6 \times \frac{2^2}{4^2} \right)}$$

$$= = \frac{-\frac{9}{4}}{3 \times \frac{4}{16}}$$

A = -3, Hence $|A| = 3$

Hence III $\rightarrow$ R

(IV) Number of maxima = $n - \ell$

So number of maxima for $4p_y = 4 - 1 = 3$

and number of maxima for $7d_{xy} = 7 - 2 = 5$

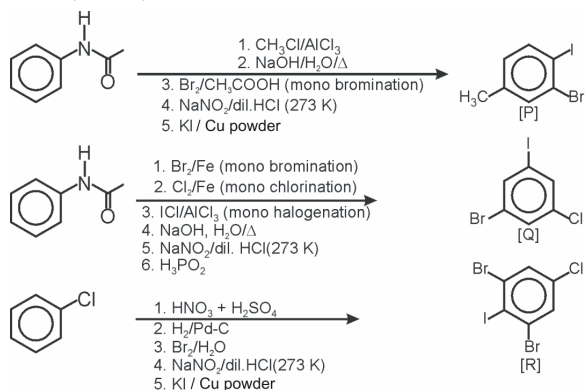
Hence same of maxima = 8

Hence IV $\rightarrow$ Q

PART-2 : CHEMISTRY

SECTION-II

1. Ans (9.00)



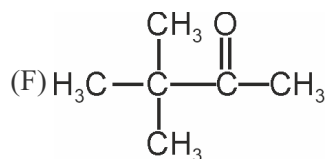
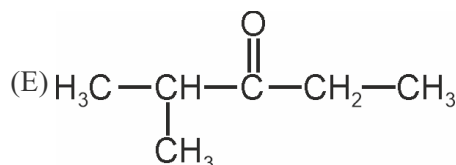
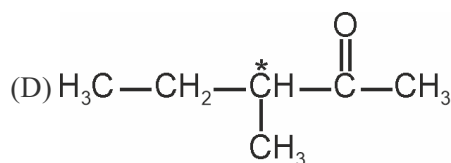
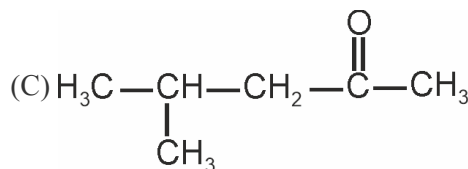
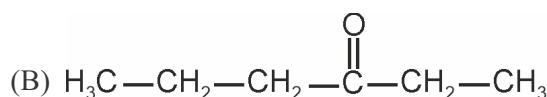
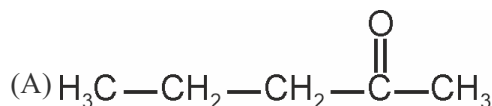
2. Ans (4.00)

I, III, VI and VIII compounds satisfy the conditions

3. Ans (2.00)

General formula of isomeric ketone of

MW = 100 is C6H12O possible structures:



Compound B and E will give racemic mixture.

5. Ans (2.50)

X = 10, Y = 4.

6. Ans (1.30)

$$\text{pH} = 7 + \frac{1}{2} [6 + \log C]$$

$$9 = 7 + \frac{1}{2} [6 + \log C]$$

$$C = 10^{-2}$$

$$C = \frac{4}{M_w \times 4} = 10^{-2}$$

$$M_w = 10^2 \text{ g.}$$

$$d = \frac{4 \times 10^2}{6 \times 10^{23} \times (800 \times 10^{-10})^3}$$

7. Ans (3.73 or 3.74)

At the end of 4 hours.

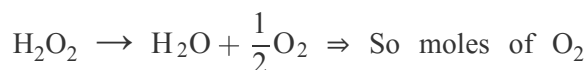
$$\text{Moles of } \text{H}_2\text{O}_2 \text{ remains} = \frac{8 \times 1}{2} = 4$$

at the end of 12th hours.

$$\text{Moles of } \text{H}_2\text{O}_2 \text{ remains} = \frac{8 \times 1}{2^3} = 1$$

Moles of H_2O_2 decomposed in the given time

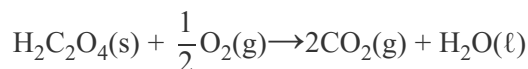
$$\text{interval} = 4 - 1 = 3$$



formed = 1.5

$$w = -P\Delta v = -nRT = \frac{-1.5 \times 8.3 \times 300}{1000} = -3.735 \text{ kJ.}$$

8. Ans (1.88 or 1.89)



$$\Delta n_g = 1.5$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta U = \Delta H - \Delta n_g RT$$

$$-300 - \frac{1.5 \times 8.3 \times 300}{1000}$$

$$-303.735 \text{ kJ/mol}$$

$$q = \frac{1.4}{90} \times 303.71 = 4.735 \text{ kJ}$$

$$\text{Heat capacity} = \frac{4.72}{2.5} \text{ kJ/K} = 1.88 \text{ or } 1.89$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C,D)

$$\therefore x = \sqrt{5 + \sqrt{5 + \sqrt{5 + \dots}}}$$

$$\Rightarrow x = \sqrt{5 + x}$$

$$\Rightarrow x^2 - x - 5 = 0$$

$$\therefore P(x) = x^2 - x - 5$$

$$\therefore \frac{1}{P(r) + 5} = \frac{1}{r^2 - r} = \frac{1}{r - 1} - \frac{1}{r}$$

$$100 \sum_{r=2}^{100} \left(\frac{1}{r-1} - \frac{1}{r} \right) = 100 \left(1 - \frac{1}{100} \right) = 99$$

$$\frac{1}{P(r) + 3} = \frac{1}{r^2 - r - 2} =$$

$$\frac{1}{3} \left[\frac{1}{(r-2)} - \frac{1}{r+1} \right]$$

$$\therefore \frac{1}{3} \sum_{r=3}^n \left(\frac{1}{r-2} - \frac{1}{r+1} \right)$$

$$= \frac{1}{3} \left[\left(\frac{1}{1} - \frac{1}{4} \right) + \left(\frac{1}{2} - \frac{1}{5} \right) + \dots + \left(\frac{1}{n-2} - \frac{1}{n+1} \right) \right]$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{3} \sum_{r=3}^n \left(\frac{1}{r-2} - \frac{1}{r+1} \right) =$$

$$\frac{1}{3} \left(1 + \frac{1}{2} + \frac{1}{3} \right) = \frac{11}{18}$$

$$\text{and } \int_0^1 \frac{1}{P(x) + x + 6} dx = \int_0^1 \frac{1}{x^2 + 1} dx$$

$$= (\tan^{-1} x)_0^1 = \frac{\pi}{4}$$

$$\text{and } \int_1^2 \frac{1}{\sqrt{P(x) + x + 4}} dx = \int_1^2 \frac{1}{\sqrt{x^2 - 1}} dx$$

$$= \left(\ln \left(x + \sqrt{x^2 - 1} \right) \right)_1^2 = \ln (2 + \sqrt{3})$$

2. Ans (B,C)

$$\begin{aligned} \therefore R_n &= \sum_{r=1}^{2n} x^{4r-3} = x + x^5 + x^9 \dots\dots\dots \\ S_n &= \sum_{r=1}^{2n} x^{4r-2} = x^2 + x^6 + x^{10} \dots\dots\dots \\ T_n &= \sum_{r=1}^{2n} x^{4r-1} = x^3 + x^7 + x^{11} \dots\dots\dots \\ U_n &= \sum_{r=1}^{2n} x^{4-r} = x^4 + x^8 + x^{12} \dots\dots\dots \end{aligned}$$

clearly, $R_n > S_n > T_n > U_n$ as $x \in (0, \frac{\pi}{4})$

$$\begin{aligned} \text{and } \lim_{n \rightarrow \infty} (R_n + S_n + T_n + U_n) \\ = x + x^2 + \dots\dots\dots = \frac{x}{1-x} \end{aligned}$$

$$\begin{aligned} \therefore R_n + S_n &= T_n + U_n \\ \Rightarrow \frac{x(1-x^{8n})}{1-x^4} + \frac{x^2(1-x^{8n})}{1-x^4} \\ &= \frac{x^3(1-x^{8n})}{1-x^4} + \frac{x^4(1-x^{8n})}{1-x^4} \\ \Rightarrow x + x^2 &= x^3 + x^4 \\ \Rightarrow (x+x^2)(x^2-1) &= 0 \\ \Rightarrow \text{no value of } x \end{aligned}$$

3. Ans (A,B,D)

(A) We have $\alpha + \beta = \alpha^2 + \beta^2$ and $\alpha\beta = \alpha^2\beta^2$
 $\therefore$ Number of quadratic equations with real roots are 3. Ans

(B) We have
 $\tan 3\theta = \tan(\pi - 2\theta)$
 $\Rightarrow 3\theta = n\pi + (\pi - 2\theta), n \in I \Rightarrow \theta = (n+1) \frac{\pi}{5}, n \in I$

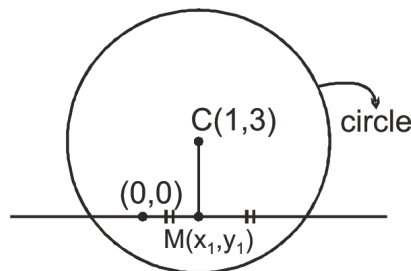
Putting $n = -1, 0, 1, 2, 3, 4$
 we get $\theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi$

$\therefore$ Number of solutions of the equation in the interval $[0, \pi] = 6$. Ans.

(C) Using A.M. $\geq$ G.M. (for positive numbers)

$$\text{So, } \frac{\frac{2x_1}{x_2} + \frac{128x_3^2}{x_2^2} + \frac{x_2^3}{4x_1x_3^2}}{3} \geq 4$$

$\Rightarrow$ Minimum value = 12 Ans.



(D) Let $M(x_1, y_1)$ be the mid-point of chord.

$$m_1 m_2 = -1 \Rightarrow \left(\frac{y_1 - 3}{x_1 - 1} \right) \left(\frac{y_1}{x_1} \right) = -1$$

$\Rightarrow$ Locus of $M(x_1, y_1)$ is $x^2 + y^2 - x - 3y = 0$.

4. Ans (A,C,D)

E_1 : prime number (2, 3, 5)

E_2 : composite number (4, 6)

E_3 : neither prime nor composite (1)

$$P(A \text{ wins}) = P(E_1) + P(\bar{E}_1 \bar{E}_2 \bar{E}_3 E_1) + \dots\dots\dots$$

$$\begin{aligned} &= \frac{1}{2} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{5}{6} \cdot \frac{1}{2} + \left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{5}{6} \right)^2 \cdot \frac{1}{2} + \dots\dots\dots \\ &= \frac{\frac{1}{2}}{1 - \frac{5}{18}} = \frac{9}{13} \end{aligned}$$

$$\begin{aligned} P(B \text{ wins}) &= P(\bar{E}_1 E_2 \text{ or } \bar{E}_1 \bar{E}_2 \bar{E}_3 \bar{E}_1 E_2 \text{ or } \dots\dots\dots) \\ &= \frac{1}{2} \cdot \frac{1}{3} + \left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{5}{6} \cdot \frac{1}{2} \right) \cdot \frac{1}{3} + \left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{5}{6} \cdot \frac{1}{2} \right)^2 \cdot \frac{1}{3} + \dots\dots\dots \\ &= \frac{\frac{1}{6}}{1 - \frac{5}{18}} = \frac{3}{13} \end{aligned}$$

$$P(C \text{ wins}) = 1 - \frac{9}{13} - \frac{3}{13} = \frac{1}{13}$$

5. Ans (B,C)

$$\vec{r} \times (\vec{p} \times \vec{q}) = \vec{p} \times (\vec{p} \times \vec{q})$$

$$\Rightarrow (\vec{r} - \vec{p}) \times (\vec{p} \times \vec{q}) = 0$$

$$\Rightarrow \vec{r} = \vec{p} + \lambda (\vec{p} \times \vec{q})$$

$$[\vec{p} \vec{q} \vec{r}] = 0 + \lambda |\vec{p} \times \vec{q}|^2 \Rightarrow 12 = \lambda \cdot 9 \Rightarrow \lambda = \frac{4}{3}$$

$$\therefore \vec{r} = \vec{p} + \frac{4}{3} (\vec{p} \times \vec{q})$$

$$|\vec{r}|^2 = |\vec{p}|^2 + \frac{16}{9} |\vec{p} \times \vec{q}|^2 + \frac{8}{3} \vec{p} \cdot (\vec{p} \times \vec{q})$$

$$= 4 + \frac{16}{9} \times 9 + 0 = 20$$

$$\therefore |\vec{r}| = 2\sqrt{5}$$

$$|\vec{r} - \vec{p}| = \lambda |\vec{p} \times \vec{q}| = \frac{4}{3} \times 3 = 4 \Rightarrow (B)$$

$$\begin{aligned} (\vec{r} - \vec{p}) \cdot \vec{r} &= \lambda (\vec{p} \times \vec{q}) \cdot \vec{r} \\ &= \lambda [\vec{p} \vec{q} \vec{r}] = \frac{4}{3} \times 12 = 16 \Rightarrow (C) \end{aligned}$$

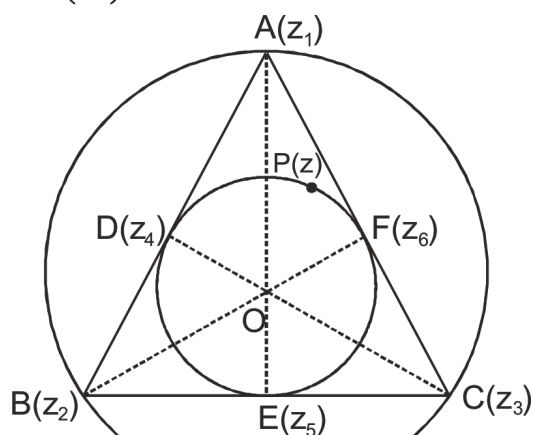
6. Ans (A, C)

$$\begin{aligned} & \int_0^{\frac{\pi}{4}} \left(t - \frac{\pi}{4}\right)^{100} \sec^2 t \, dt = \left(\left(t - \frac{\pi}{4}\right)^{100} \tan t \right)_0^{\frac{\pi}{4}} - \\ & \int_0^{\frac{\pi}{4}} 100 \left(t - \frac{\pi}{4}\right)^{99} \tan t \, dt \\ & \frac{\int_0^{\frac{\pi}{4}} \left(t - \frac{\pi}{4}\right)^{100} \sec^2 t \, dt}{\int_0^{\frac{\pi}{4}} \left(t - \frac{\pi}{4}\right)^{99} \tan t \, dt} = -100 = I_1 \\ I_2 &= \frac{1}{100} \left(\int_0^{\frac{\pi}{4}} \left(t - \frac{\pi}{4}\right)^{100} \sec^2 t \, dt - 100 \int_0^{\frac{\pi}{4}} \left(t - \frac{\pi}{4}\right)^{99} \tan t \, dt \right) \\ &= \frac{1}{100} \int_0^{\frac{\pi}{4}} \left(t - \frac{\pi}{4}\right)^{100} \sec^2 t - 100 \int_0^{\frac{\pi}{4}} \left(\frac{\pi}{4} - t\right)^{99} \tan t \, dt \\ &= \frac{1}{100} \int_0^{\frac{\pi}{4}} \left(\left(t - \frac{\pi}{4}\right)^{100} \sec^2 t + 100 \left(t - \frac{\pi}{4}\right)^{99} \tan t \right) dt \\ &= \frac{1}{100} \left(\left(t - \frac{\pi}{4}\right)^{100} \tan t \right)_0^{\frac{\pi}{4}} = 0 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)



$$\begin{aligned} \text{(I)} \quad & \because |z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\operatorname{Re}(z_1 \bar{z}_2) \\ & \Rightarrow (2\sqrt{3})^2 = 2^2 + 2^2 - 2\operatorname{Re}(z_1 \bar{z}_2) \\ & \Rightarrow \operatorname{Re}(z_1 \bar{z}_2) = -2 \end{aligned}$$

Similarly, $\operatorname{Re}(z_2 \bar{z}_3) = -2$

& $\operatorname{Re}(z_3 \bar{z}_1) = -2$

$$\therefore \operatorname{Re}(z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1) = -6$$

$$\begin{aligned} \text{(II)} \quad & \because \frac{z_1 - 0}{z_3 - 0} = e^{i\frac{2\pi}{3}} \Rightarrow \frac{z_1}{z_3} = \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) \\ & \Rightarrow \frac{4z_1}{z_3} = 2(-1 + i\sqrt{3}) \end{aligned}$$

$$\therefore a = 2$$

$$\begin{aligned} \text{(III)} \quad & |z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2 \\ &= 2(|z_1|^2 + |z_2|^2 + |z_3|^2) + 2\operatorname{Re}(z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1) \end{aligned}$$

$$= 2(4 + 4 + 4) + 2(-6) = 12$$

$$\begin{aligned} \text{(IV)} \quad & DP^2 + EP^2 + FP^2 = |z - z_4|^2 + |z - z_5|^2 + \\ & |z - z_6|^2 = 3|z|^2 + |z_4|^2 + |z_5|^2 + |z_6|^2 = 6 \end{aligned}$$

8. Ans (D)

$$\text{(I)} \quad \because I = \int_0^a \ln(\cot a + \tan x) dx \quad \dots (1)$$

using king

$$\begin{aligned} I &= \int_0^a \ln(\cot a + \tan(a - x)) dx \\ &= \int_0^a \ln\left(\cot a + \frac{\tan a - \tan x}{1 + \tan a \tan x}\right) dx \\ &= \int_0^a \ln\left(\frac{\cot a + \tan a}{1 + \tan a \tan x}\right) dx \quad \dots (2) \end{aligned}$$

From (1) + (2) we get

$$2I = \int_0^a \ln\left(\frac{\cot a + \tan a}{\tan a}\right) dx = a \ln \operatorname{cosec}^2 a$$

$$I = a \ln(\operatorname{cosec} a)$$

(II) $\because$ Given expression

$$\begin{aligned} &= (1 + x + x^2 + x^3 + x^4)^{1001} (1 - x)^{1002} \\ &= \left(\frac{1 - x^5}{1 - x}\right)^{1001} (1 - x)^{1002} \\ &= (1 - x)(1 - x^5)^{1001} \end{aligned}$$

$\therefore$ term in the expansion will be either of x^{5m} or x^{5m+1} form, where $m \in \mathbb{N}$

$$\therefore \text{Coefficient of } x^{2017} = 0$$

(III) Hyperbola be $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$

$$B^2 = A^2(e^2 - 1) = a^2e^2 - A^2 = (a^2 - b^2) - A^2$$

$$\therefore \text{Hyperbola is } \frac{x^2}{A^2} + \frac{y^2}{A^2 - (a^2 - b^2)} = 1$$

$$\therefore \left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}} \right) \text{ lies on it}$$

$$\therefore \frac{a^2}{2A^2} + \frac{b^2}{2(A^2 - a^2 + b^2)} = 1$$

$$\Rightarrow a^2(A^2 - a^2 + b^2) + b^2A^2$$

$$= 2A^2(A^2 - a^2 + b^2)$$

$$\Rightarrow 2A^4 + A^2(b^2 - 3a^2) + a^4 - a^2b^2 = 0$$

$$\Rightarrow A^2 = \frac{(3a^2 - b^2) \pm \sqrt{(b^2 - 3a^2)^2 - 4 \cdot 2 \cdot (a^4 - a^2b^2)}}{4}$$

$$\Rightarrow A^2 = \frac{(3a^2 - b^2) \pm (a^2 + b^2)}{4}$$

$$= a^2 \text{ or } \frac{a^2 - b^2}{2}$$

$$\therefore A \neq a$$

$$\therefore A^2 = \frac{a^2 - b^2}{2}$$

$$\therefore B^2 = \frac{a^2 - b^2}{2}$$

$$\therefore e = \sqrt{2} \Rightarrow e^2 = 2$$

$$(IV) \therefore \left\{ \frac{1}{x} \right\} = \{x^2\}$$

$$\Rightarrow \frac{1}{x} = x^2 - 2 \quad \therefore x \in (\sqrt{2}, \sqrt{3})$$

$$\Rightarrow x^3 - 2x - 1 = 0$$

$$\Rightarrow (x + 1)(x^2 - x - 1) = 0$$

$$\therefore x \neq -1 \quad \therefore x^2 - x - 1 = 0$$

$$\Rightarrow x^2 = x + 1 \Rightarrow x^4 = x^2 + 2x + 1$$

$$\Rightarrow x^4 = 3x + 2$$

$$\therefore x^4 - \frac{3}{x} = 3x + 2 - 3(x^2 - 2)$$

$$= 8 - 3(x^2 - x)$$

$$= 8 - 3(1) = 5$$

9. Ans (B)

(I) CS_1 and CS_2 divide the triangle into three of equal area so that S_1 and S_2 are points of trisection of AB.

Thus $S_1 = (0, 0)$, $S_2 = (2, -3)$.

Equation of CS_1 is $y - x = 0$.

Slope of $CS_2 = -4$.

The line through $(0, 0)$ drawn parallel to

CS_2 is $y + 4x = 0$

Required equation is $y^2 + 3xy - 4x^2 = 0$

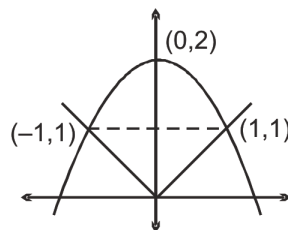
$$\Rightarrow \lambda + \mu = 3 + 4 = 7 \text{ Ans.}$$

(II) The required area is bounded by the lines $y = \pm x$ and the parabola $x^2 = -(y - 2)$ having the vertex at $(0, 2)$ and passing through $(\pm 1, 1)$.

By symmetry about the y-axis,

$$\text{Area} = 2 \left[\frac{1}{2} + \int_1^2 \sqrt{2-y} dy \right] = \frac{7}{3} \equiv \frac{p}{q}$$

$$\Rightarrow p - q = 7 - 3 = 4 \text{ Ans.}$$



(III) With $y = xv$,

$$xv' = \frac{1+v^2}{2v} - v = \frac{1-v^2}{2v} \Rightarrow \frac{2v dv}{1-v^2} = \frac{dx}{x}$$

whose solution is $x(1-v^2) = c$.

When $x = 2$, $v = \frac{1}{2}$ so that $c = \frac{3}{2}$.

The equation of the curve is $x^2 - y^2 = \frac{3x}{2}$

$$\Rightarrow \left(x - \frac{3}{4} \right)^2 - y^2 = \frac{9}{16}$$

This is a rectangular hyperbola with eccentricity $\sqrt{2}$.

$$\therefore e^4 = 4 \text{ Ans.}$$

$$\begin{aligned} \text{(IV)} \quad P &= \lim_{n \rightarrow \infty} \frac{[(n^2 + 1^2)(n^2 + 2^2) \dots (n^2 + n^2)]^{m/n}}{n^{2m}} \\ \ln P &= \lim_{n \rightarrow \infty} \frac{m}{n} \left[\sum_{r=1}^n \ln(n^2 + r^2) - 2m \ln n \right] \\ \ln P &= m \lim_{n \rightarrow \infty} \left[\sum_{r=1}^n \frac{\ln(n^2 + r^2) - 2n \ln n}{n} \right] \\ \ln P &= m \left[\ln 2 + \frac{\pi}{2} - 2 \right] \\ P &= e^{m \left[\ln 2 + \frac{\pi}{2} - 2 \right]} = e^{m \ln 2} \cdot e^{-2m} \cdot e^{\frac{m\pi}{2}} \\ &= \frac{2^m e^{\frac{m\pi}{2}}}{e^{2m}} = \frac{(2e^{\pi/2})^m}{e^{2m}} = \left(\frac{2\sqrt{e\pi}}{e^2} \right)^m \\ \Rightarrow a &= 2 \quad \text{Ans.} \end{aligned}$$

10. Ans (C)

(I) Equation of the plane

$$A(x-1) + B(y-1) + C(z-1) = 0 \quad \dots(1)$$

Since the line is perpendicular to the plane (1)

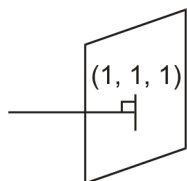
$$\therefore 3(x-1) + 0(y-1) + 4(z-1) = 0$$

$$3x + 0y + 4z - 7 = 0$$

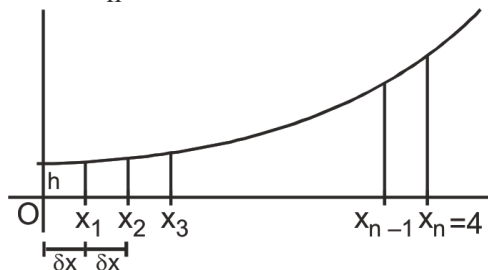
distance from $(0, 0, 0)$

$$d = \left| \frac{-7}{5} \right| = \frac{7}{5} = \frac{p}{q}$$

$$\Rightarrow p - q = 2 \text{ Ans}$$



$$\begin{aligned} \text{(II)} \quad L &= \lim_{\delta x \rightarrow 0} \delta x (x_1 + x_2 + x_3 + \dots + x_n) \\ &= \lim_{n \rightarrow \infty} \frac{4}{n} \left[\frac{4}{n} + \frac{8}{n} + \frac{12}{n} + \dots + 4 \cdot \frac{n}{n} \right] \\ (\delta x &= \frac{4}{n}) \\ &= \lim_{n \rightarrow \infty} \frac{16}{n^2} (1 + 2 + 3 + \dots + n) \\ &= \lim_{n \rightarrow \infty} \frac{16}{n^2} \cdot \frac{n(n+1)}{2} = 8 \text{ Ans.} \end{aligned}$$



10. Ans (C)

(III) If x is replaced by $-x$ in the given equation then

$$-x f(-x) + (1+x) f(x) = x^2 - x + 1$$

subtracting the two equations we get

$$f(-x) = f(x) + 2x,$$

substituting the value of $f(-x)$ in the given equation
we get

$$xf(x) + (1 - x)(f(x) + 2x) = x^2 + x + 1 \text{ and thus}$$

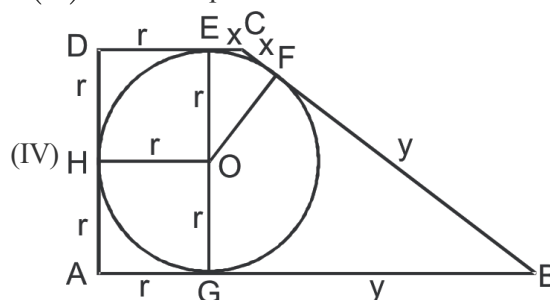
$$f(x) = x^2 + x + 1 - 2x(1 - x)$$

$$= 3x^2 - x + 1 = 3 \left(x - \frac{1}{6} \right)^2 + \frac{11}{12}$$

and hence $f(x) \geq \frac{11}{12}$.

If $x = \frac{1}{6}$ then we get

$$f\left(\frac{1}{6}\right) = \frac{11}{12} \equiv \frac{p}{q} \Rightarrow (q-p) = 1 \text{ Ans.}$$



Using the notations from the above diagram and the conditions from the problem one obtains :

$$3(r+x) = r+y \Leftrightarrow y = 3x + 2r$$

$$\text{and } (x+y)^2 = 4r^2 + \{(r+y)-(r+x)\}^2 \Leftrightarrow xy = r^2$$

Hence $x(3x + 2r) = r^2 \Leftrightarrow x = \frac{r}{3}$

and consequently $y = 3r$. On the other hand the area of the trapezoid ABCD is 4, thus

$$(r + x + r + y)r = 4$$

Substituting for $x = \frac{r}{3}$

and $y = 3r$ we get $r = \frac{\sqrt{3}}{2}$

$$\Rightarrow 4r^2 = 3 \text{ Ans.}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3.00)**

$$a + b + 4c < 41$$

$$\Rightarrow a + b + 4c + t = 40$$

$\Rightarrow$ Beggar's method $a, b, c \in \mathbb{N}$

$$\Rightarrow x + y + 4c + t = 38$$

$$- - 4 - = 34 \Rightarrow {}^{36}C_2$$

$$- - 8 - = 30 \Rightarrow {}^{32}C_2$$

$$: : : : = : :$$

$$- - : - = : :$$

$$\Rightarrow \frac{{}^{36}C_2 + {}^{32}C_2 + {}^{28}C_2 + {}^{24}C_2 + \dots + {}^4C_2}{730}$$

$$= \frac{\sum_{r=1}^9 {}^{4r}C_2}{730} = \frac{\sum_{r=1}^9 \frac{4r(4r-1)}{2}}{730}$$

$$= \frac{8 \sum_{r=1}^9 r^2 - 2 \sum_{r=1}^9 r}{730}$$

$$= \frac{\frac{8 \times 9 \times 10 \times 19}{6} - \frac{2 \times 9 \times 10}{2}}{730}$$

$$= \frac{2280 - 90}{730} = \frac{2190}{730} = 3.$$

2. **Ans (3.00)**

$$\frac{b+c}{2} = a \text{ and } b = \sqrt{ac}$$

$$\therefore 2a - c = \sqrt{ac} \Rightarrow 4a^2 + c^2 - 5ac = 0$$

$$\Rightarrow (4a - c)(a - c) = 0$$

$$\therefore a, b \text{ and } c \text{ are distinct } \therefore c \neq a$$

$$\therefore c = 4a \text{ \{both negative, given\}}$$

$$\therefore b = +\sqrt{ac} = \sqrt{4a^2} = 2|a| = -2a$$

$\{ \because b > 0 \text{ is given} \}$

$$\therefore \text{The G.P. will be } a, -2a, 4a$$

$$\therefore \text{Common ratio} = -\frac{2a}{a} = -2 = r.$$

$$I = \int_{-1}^1 (x^5 + 3x^2 + \operatorname{sgn}[x] + [x] + 1) dx$$

$$= 2 \int_0^1 (3x^2 + 1) dx + \int_{-1}^1 [x] dx$$

$$= 2 \left[x^3 + x \right]_0^1 + \int_{-1}^0 [x] dx + \int_0^1 [x] dx$$

$$= 2(2) + (-1) + 0 = 3.$$

3. **Ans (4.00)**

Let degree of $f(x)$ be n .

$$\therefore \text{degree of } f(f(f(x))) = \text{degree of } f(x)$$

$$\Rightarrow n^3 = n \Rightarrow n = 1$$

$$\therefore \text{Let } f(x) = 2x + b \text{ (} \because \text{leading coefficient} = 2)$$

$$\therefore f(f(f(x))) = 2(4x + 3b) + b = 8x + 7b$$

$$\therefore 8 + 2 - 2p = 0 \Rightarrow p = 5$$

$$\text{and } 7b - 4b = 3 \Rightarrow b = 1$$

$$\therefore f(x) = 2x + 1$$

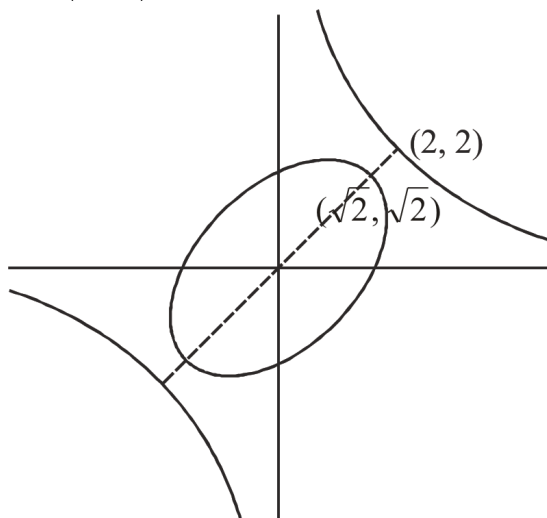
$$f(f(x)) = 4x + 3$$

$$\therefore \frac{d}{dx} f(f(x)) = 4$$

$$f^{-1}(x) = \frac{x-1}{2} \Rightarrow 2f^{-1}(x) + 1 = x$$

$$\therefore \int_{-p}^p (2f^{-1}(x) + 1) dx = 0$$

4. Ans (4.00)



Clearly P lies on given ellipse and Q (2tan C, 2cot C) lies on rectangular hyperbola $xy = 4$

$$\therefore \text{Minimum value} = (2\sqrt{2} - 2)^2$$

$$= 12 - 8\sqrt{2} \equiv a + b\sqrt{2}$$

$$\therefore a + b = 4. \text{ Ans.}$$

5. Ans (8.00)

$\therefore$ Elements of determinant can be 1 or -1.

$\therefore$ First two places of first row can be filled in $2 \times 2 = 4$ ways and last place can be filled in only one way to product be 1.

$\therefore$ First row can be made in 4 ways. similarly second row can be made in 4 ways. and third row can be made in 1 way only.

$$\therefore \text{Number of elements in } S = 16 = m$$

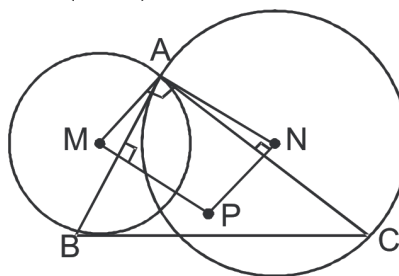
$$\therefore |P| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1(-2 + 3) = 1$$

$$\therefore \text{adj}(\text{adj } P) = |P|^{n-2} \cdot P = P$$

$$\Rightarrow \text{tr}(\text{adj}(\text{adj } P)) = \text{tr}(P) = 2 = n$$

$$\therefore \frac{m}{n} = 8$$

6. Ans (2.00)



clearly PM is perpendicular bisector of AB

and PN is perpendicular bisector of AC

$\therefore$ P is circumcentre of $\triangle ABC$

$$\therefore PA = PB = PC = R \text{ (circum radius)}$$

$$\therefore a + b + c = 2R \sin A + 2R \sin B + 2R \sin C$$

$$\therefore K = 2$$

7. Ans (8.00)

$$xy + 2z = \left(\log_2 3 - 4 \log_{\left(\ln \frac{5}{4}\right)} 3 \right) 2 \log_3 \left(\ln \frac{5}{4} \right)$$

$$+ 2 \log_2 \left(\log_{\frac{5}{4}} e \right)$$

$$= 2 \log_2 \left(\ln \frac{5}{4} \right) - 8 - 2 \log_2 \left(\ln \frac{5}{4} \right) = -8$$

$$\therefore |xy + 2z| = 8$$

8. Ans (5.00)

$$\therefore \tan 30^\circ = \frac{3 \tan 10^\circ - \tan^3 10^\circ}{1 - 3 \tan^2 10^\circ} = \frac{1}{\sqrt{3}}$$

$$3(3 \tan 10^\circ - \tan^3 10^\circ)^2 = (1 - 3 \tan^2 10^\circ)^2$$

$$\Rightarrow 3(9 \tan^2 10^\circ + \tan^6 10^\circ - 6 \tan^4 10^\circ)$$

$$= 1 + 9 \tan^4 10^\circ - 6 \tan^2 10^\circ$$

$$\Rightarrow 3 \tan^6 10^\circ - 27 \tan^4 10^\circ + 33 \tan^2 10^\circ = 1$$

$$\text{Now, } \log_2 [x - 2] > 1$$

$$[x - 2] > 2 \Rightarrow [x - 2] \geq 3$$

$$x - 2 \geq 3 \Rightarrow x \geq 5.$$